KINETC: Koopman-Inspired Nonlinear Event-Triggered Control from Data

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***Abstract*—Event-triggered Control (ETC) presents a promising paradigm for efficient resource usage in networked and embedded control systems by reducing communication instances compared to traditional time-triggered strategies. This paper introduces a novel approach to ETC for discrete-time nonlinear systems using a data-driven framework. By leveraging Koopman operator theory, the nonlinear system dynamics are globally linearized (approximately) in a higher-dimensional space. We design a state-feedback controller and an event-triggering policy directly from data, ensuring exponential stability in Lyapunov sense. The proposed Koopman-Inspired Nonlinear Event-Triggered Control from Data (KINETC) method is validated through extensive simulation experiments, demonstrating significant resource savings by reducing the communication instances by 40%.**

I. Introduction

ETC is an implementation strategy in which the plant and its controller only exchange data when certain output- or state-related conditions are met. Event-triggered control seeks to reduce communication instances by concentrating on the real needs of the system, in contrast to traditional conservative time-triggered strategies that depend on fixed time intervals for communication. In situations where efficient use of resources is essential, like networked control systems and embedded systems, this paradigm has become more and more popular. ETC strategies, which offer improved system performance and resource savings in a variety of setups and control problems, have been developed in the literature thanks to the early results by [1, 2] and the work of Tabuada [3], and Heemls et al. [4].

In the most of the current research, parametric state-space models are the the foundation of traditional control engineering, where the plant to be controlled must first be identified or modeled. These models make use of the data from the system (from the input and the output) and are frequently written down starting from physics first principles or architecturally constrained system identification techniques. But in cases when first-principles models are intricate or hard to derive, they are only approximate representations of real systems, which inevitably leads to modeling errors. These errors impede accurate control design and spread through the phases of analysis and implementation, affecting the overall performance of the system.

By excluding the demand for explicit system identification and instead of leveraging data gathered from open-loop simulations/experiments for any system control analysis and design, data-driven control techniques serves as a promising alternative. Several data-driven techniques for creating state feedback controllers and illustrating system dynamics have been shown in recent works, such as those by da Silva et al. [5], and De Persis and Tesi [6]. These techniques greatly streamline the control design process and do not require continuously exciting input data. There also exist numerous application of data-driven control in fields like robotics [7, 8], aerospace [9], and in power systems [10].

However, an alternative option in the situation, when the first principles or system identification techniques fail, would be to construct the controller directly using the input, state, and output data that is now accessible. This approach is known as direct data-driven control [11–13]. Although, the literature is full of data-driven techniques for control, only a limited number of techniques exist in the current literature for data-driven event-based control, particularly for nonlinear systems such as [14–16].

There exists, therefore, a profound demand for comprehensive data-driven event-based control methods tailored for general nonlinear systems, particularly applicable to discrete-time systems in our case. In many cases, it is appropriate and feasible to formulate the control and triggering conditions as data-dependent Linear Matrix Inequality (LMI). Given that most of the existing literature on ETC is well-developed for Linear Time Invariant (LTI) systems, we aim to globally linearize nonlinear systems by increasing the dimensional space in which they reside.

This is not totally a new idea, as Koopman [17] in 1931 presented a trade-off between the nonlinear nature of dynamical systems and infinite dimensional representation of the same nonlinear system but it will look linear in the lifted space. Koopman idea has been recently revisited in many research fields [18–21] due to the intersection between data science and easy to access computational domain.

We consequently propose KINETC, an idea inspired by Koopman Operator (KO) to acquire (approximately) global linear systems but in a higher dimensional space. Afterwards, we design the controller and the triggering policy for ETC for discrete-time linear systems directly from data, all together ensuring performance metrics (i.e. exponential stability in Lyapunov sense).

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II. Preliminaries

A. Notation

Let $\mathbb{Z}_{\geq 0} := \{0, 1, 2, \dots\}$ denotes the set of non-negative integers and $\mathbb{Z}_{>0} := \mathbb{Z}_{\geq 0} \backslash \{0\}$. We denote $\mathbb{R}$ to be the set of real numbers and follows the same notation as in $\mathbb{Z}$. The L_2 norm of a vector is denoted $\|\cdot\|$. The notation I and 0 denote the identity and the zero matrix, respectively. Given a symmetric matrix A, the notation $A \succ 0$ means that A is positive definite, while $A \succeq 0$ means that A is positive semi-definite. Similarly, $A \prec 0$ indicates that A is negative definite, and $A \preceq 0$ indicates that A is negative semi-definite. For any matrix A, $A^\top$ denotes the transpose of A. The symbol $\mathcal{N}(\mu, \sigma^2)$ represents a standard normal distribution with mean μ and variance σ^2. The symbol λ_i denotes an eigen value of a matrix.

B. Problem Overview

Consider the discrete time dynamical system

$$x_{k+1} = f(x_k, u_k), \tag{1}$$

where $x_k \in \mathbb{R}^n$ is the state and $u_k \in \mathbb{R}^m$ is the control input, each at time instant $k \in \mathbb{Z}_{\geq 0}$ with $n, m \in \mathbb{Z}_{>0}$, and f is a transition map such that $f : \mathbb{R}^n \times \mathbb{R}^m \mapsto \mathbb{R}^n$, generally nonlinear, and assumed to be unknown, and stabilizable.

We consider a scenario where the system in (1) is connected to a controller via networked medium. Especially, the state measurements are sent to the controller via digital channel and the controller access the actuators directly. The goal is to design a data-driven event-triggered state-feedback controller with gain $K \in \mathbb{R}^{m \times n}$ to stabilize the plant in (1) while abiding by a triggering policy that defines the instances $\{k_i\}_{i \in \mathcal{Z}}$ at which a transmission happens, with $\mathcal{Z} \subseteq \mathbb{Z}_{\geq 0}$. At the time instant $k = 0$, we consider a transmission happens, so that $k_0 = 0$. In our settings, the controller is updated only upon the violation of some well defined triggering policy in contrary with the nominal Time-triggered Control (TTC). The sequence $\{k_i\}_{i \in \mathcal{Z}}$ leads to aperiodic updates of the controller. The controller then follows a zero-order hold implementation that takes the form of

$$u_k = K x_{k_i}, \quad k \in [k_i,\, k_{i+1}[. \tag{2}$$

The zero-order hold nature of the presented controller is taken into consideration by the state error

$$e_k = x_{k_i} - x_k, \tag{3}$$

which can be interpreted as the deviation between the current state and the last time an event happened. We consider the relative thresholding metric and events occur with the violation of the condition

$$\|e_k\| \leq \gamma \|x_k\| \tag{4}$$

where $\gamma > 0$ is a relative parameter for the thresholding policy. The policy in (4) is checked every time instant k, and only upon the violation of the policy, the control updates.

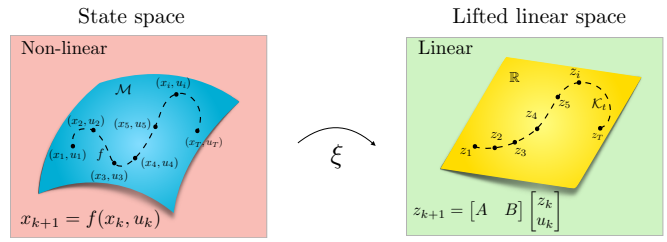

Fig. 1: Illustration of the Koopman Operator: The red panel represents the generic nonlinear state-space. Conversely, the green panel represents the linear space.

C. Persistence of Excitation

Consider a carried out experiment for the system in (1) and its states and input data are recorded in the following way

$$\mathcal{D} := \{x_k, u_k : k \in [0, (T-1)] \cap \mathbb{Z}_{\geq 0}\},$$

where T is the final time of the experiment. Let set $\mathcal{D}$ exists. Then, we define

$$\Gamma_0 := \begin{bmatrix} u_0 & u_1 & \dots & u_{T-1} \end{bmatrix} \in \mathbb{R}^{m \times T}, \tag{5a}$$

$$X_0 := \begin{bmatrix} x_0 & x_1 & \dots & x_{T-1} \end{bmatrix} \in \mathbb{R}^{n \times T}, \tag{5b}$$

$$X_1 := \begin{bmatrix} x_1 & x_2 & \dots & x_T \end{bmatrix} \in \mathbb{R}^{n \times T}. \tag{5c}$$

Assumption 1: Considering $T \geq n + m$, the matrix $\left[\begin{smallmatrix} X \\ \Gamma \end{smallmatrix}\right]$ has full row rank. □

Assumption 1 can be verified numerically for a given set $\mathcal{D}$. The results of Willems et al. [22] ensures, for discrete-time systems, the validity of assumption 1 as long as u is persistently exciting signal.

III. Framework

A. Koopman operator theory

Definition 1 (KO): Consider the system given in (1). The KO $\mathcal{K}_t$ is an infinite dimensional operator

$$\mathcal{K}_t \xi = \xi \circ f(x_k, u_k).$$

acts on $\xi \in \mathcal{H} : \mathbb{R}^n \times \mathbb{R}^m \mapsto \mathbb{R}$, the observable functions of the state space, where $\circ$ is the function composition. □

The KO acts on the Hilbert space $\mathcal{H}$ of all scalar measurement functions ξ and is by definition a *linear operator*–since for $\xi_1, \xi_2 \in \mathcal{H}$ and $\beta_1, \beta_2 \in \mathbb{R}$

$$\begin{aligned} \mathcal{K}_t(\beta_1 \xi_1, \beta_2 \xi_2) &= \beta_1 \xi_1 \circ f + \beta_2 \xi_2 \circ f \\ &= \beta_1 \mathcal{K}_t \xi_1 + \beta_2 \mathcal{K}_t \xi_2, \end{aligned} \tag{6}$$

An infinite-dimensional space $\mathcal{H}$ of observable functions is used to represent a nonlinear system linearly using this method [23]. This means that the dynamics are changed from nonlinear, finite-dimensional to linear, infinite-dimensional upon transitioning from the state-space model to the Koopman representation (see Fig. 1). Nonetheless, we are interested in a finite-dimensional approximation of the KO from a practical perspective. For those who may be interested, please refer to the following refs. [24, 25].

Also, KO provides (approximately, in a practical sense) global linear representation for nonlinear dynamics if the right,

in a sense made precise in the sequel, set of observable functions is chosen. Generally speaking, the lifting observable functions are hard to identify. They can be found either by brute-force trial and error in a specific basis for the Hilbert space (e.g., trying numerous polynomial functions, or Fourier basis functions) or by prior knowledge about the system. Several efforts have been done in that matter [26–28].

Motivated by the preceding analysis, we will make use of the idea of lifting the nonlinear dynamics from its state-space to look linear in a higher-dimensional state-space.

Remark 1: Our method necessitates a blurry prior physical knowledge of the underlying plant, not necessarily a complete knowledge, but at least a knowledge that can describe the domain shape at which the system operates in to design the lifting observable functions. Several methods can help in constructing analytical/heuristic observable functions can be found in [29, 30] – just to name a few. □

Remark 2: At this stage of the work, we design the controller directly from the data. This step requires a set of observable functions that are satisfactory to approximate the KO with notes regarding that being discussed in remark 1. And sense we do not focus on the approximate identification of the KO itself, we did not include discussion on such topic. However, in more general scenarios one may need to identify the operator for any purpose. In such case, we refer the interested readers to [31, 32]. □

Now, the collected set $\mathcal{D}$ should be revised. Instead of having the system's states only, we must consider the additional observable functions taking the form of $\Xi(x) = \begin{bmatrix}\xi_1(x) & \xi_2(x) & \dots & \xi_p(x)\end{bmatrix}^\top$. The observable functions $\Xi \in \mathbb{R}^p (p > n)$. Note that we only lift the state not the control. So the set $\mathcal{D}$ becomes

$$\Gamma_0 := \begin{bmatrix}u_0 & u_1 & \dots & u_{T-1}\end{bmatrix} \in \mathbb{R}^{m\times T}, \tag{7a}$$
$$Z_0 := \Xi(X_0) \in \mathbb{R}^{p\times T}, \tag{7b}$$
$$Z_1 := \Xi(X_1) \in \mathbb{R}^{p\times T}. \tag{7c}$$

Remark 3: In response to this change, a slight modification of assumption 1, where $T \geq n + m$ will be replaced by $T \geq p + m$.

At the end of the day, after choosing the set of observable functions, the system in (1) can be now formulated as

$$z_{k+1} = Az_k + Bu_k, \tag{8a}$$
$$x_k = Cz_k \tag{8b}$$

B. Event-triggered Control For the Lifted Representation of the Non-linear Dynamics

Let's consider the system given in (8), the globally linear version of the system in (1), subject to the controller (2) that result in

$$z_{k+1} = Az_k + BKz_{k_i} \tag{9a}$$
$$= Az_k + BKz_k + BKz_{k_i} - BKz_k \tag{9b}$$
$$= (A + BK)z_k + BKe_k, \quad \forall k \in [k_i,\, k_{i+1}[, \tag{9c}$$

which can be understood as a closed-loop representation of the system in (8) with the state error.

An alternative representation of the event-triggered closed loop system should be derived to account for the data-driven nature of this work. In ref. [33] Persis et al. derived a data-driven representation of the closed loop system without considering the matrix BK. On the other hand, Digge and Pasumarthy [15] arrived to a closed loop representation that allows dealing with the event-triggered formulation. We now borrow the work of [15], with a slight modification to account for the lifted linear representation of the nonlinear dynamics.

Lemma 1 (Data-driven event-triggered cloosed loop representation [6, 15]): Let assumption 1 hold. Then the equivalent data-driven closed loop representation of the system (9) is given by

$$z_{k+1} = Z_1Lz_k + Z_1Ne_k, \tag{10}$$

where L and N a $T \times p$ matrices that satisfy

$$\begin{bmatrix} I & 0 \\ K & K \end{bmatrix} = \begin{bmatrix} Z_0 \\ \Gamma_0 \end{bmatrix} \begin{bmatrix} L & N \end{bmatrix}. \tag{11}$$

□

Proof 1: See ref. [15, Section III. B]. ■

Since a formulation is derived. We move forward to derive the condition for system (10) to be exponentially stable in Lyapunov sense. To explain further, a linear system described by $z_{k+1} = Az_k$, where $A \in \mathbb{R}^p$, is considered exponentially stable if there exists a function $V : \mathbb{R}^p \mapsto \mathbb{R}$ defined by $V(z_k) = z_k^\top Sz_k$ with $S \succ 0$, such that $V(z_{k+1}) \leq \alpha V(z_k)$ along the system's trajectories for all $k \geq 0$ and for some $\alpha \in (0, 1)$.

Considering a classical Lyaponuv candidate function descried in the later paragraph, the exponential Lyaponuv stability criteria[1] is given by

$$\begin{bmatrix} z_k \\ e_k \end{bmatrix}^\top \begin{bmatrix} L^\top Z_1^\top SZ_1L - \alpha S & L^\top Z_1^\top SZ_1N \\ N^\top Z_1^\top SZ_1L & N^\top Z_1^\top SZ_1N \end{bmatrix} \begin{bmatrix} z_k \\ e_k \end{bmatrix} \leq 0 \tag{12}$$

In this work, the design of the ETC strategy should not violate the Lyabunov stability condition in (12) to ensure exponential stability.

C. Learning Controller From Data

Firstly, we design the controller gains to stabilize the globally linearized system. We consider the data-driven closed loop representation in (10) neglecting the error at this stage

$$z_{k+1} = Z_1Lz_k, \tag{13}$$

the controller gains can be designed directly from data, following the analysis in [6, Section IV. A], by the following theorem it ensures the Lyaponuv stability condition.

Theorem 1: Let condition 1 hold. And by exploiting the results of lemma 1. Then any matrix G_1 that satisfy the following LMI,

$$\begin{bmatrix} Z_0G_1 & G_1^\top Z_1^\top \\ Z_1G_1 & Z_0G_1 \end{bmatrix} \succeq 0 \tag{14}$$

[1]The full analysis is given in the appendix.

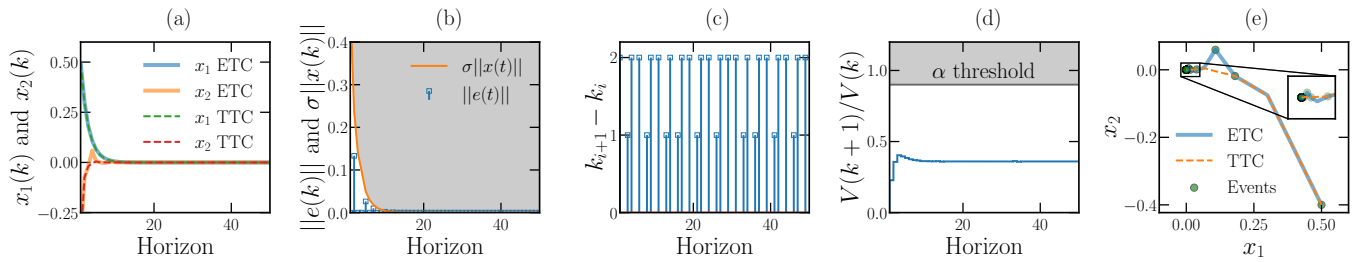

Fig. 2: Results of the illustrative example. (a) State trajectories x_1 and x_2 for both ETC and TTC over the horizon. (b) Norms of the error $\|e_k\|$ and the threshold parameter $\gamma\|x_k\|$. (c) Inter-event times $k_{i+1} - k_i$ showing the intervals between successive events. (d) The ratio of the Lyapunov function $V(x_{k+1})/V(x_k)$ with the stability threshold α. (e) The phase portrait of the system trajectory in the state7 space showing both the ETC and TTC behaviour, it also show the instances at which an event happens along the trajectory in the ETC framework.

results in

$$K = \Gamma_0 G_1 (Z_0 G_1)^{-1} \tag{15}$$

which stabilizes the system (8). □

Proof 2: To certify stability in exponential decay of the system (13) with a rate α, implies

$$L^\top Z_1 S Z_1 L - \alpha S \preceq 0, \tag{16}$$

with L satisfying (11). Let $G_1 := LS^{-1}$, and pre- and post-multiply (16) by S^{-1}, the stability of the system can be guaranteed if there exists two matrices G_1 and S such that

$$\begin{aligned} &G_1^\top Z_1^\top S Z_1 G_1 - \alpha S^{-1} \preceq 0 \\ &KS^{-1} = \Gamma_0 G_1 \\ &S^{-1} = Z_0 G_1 \end{aligned}$$

Moreover, we use $S^{-1} = Z_0 G_1$ and obtain

$$\begin{aligned} &G_1^\top Z_1^\top (Z_0 G_1) Z_1 G_1 - \alpha Z_0 G_1 \preceq 0 \\ &Z_0 G_1 \succ 0 \\ &K = \Gamma_0 G_1 (Z_0 G_1)^{-1} \end{aligned}$$

Using Schur's complement lemma on the first inequality, we reach to (14) which results in gains given from (15) that exponentially stabilize the system. ■

D. Learning the Triggering Policy from Data

In the interval $[k_i, k_{i+1}[$, it is essential that inequality (12), which ensures exponential convergence, is also satisfied. The following theorem derives a window for the parameter γ that ensures the stability of system (10).

Theorem 2: Assume that the condition 1 is satisfied. So, the relative threshold parameter γ for the event-triggered implementation (4) with the controller (15) can be calculated by solving for γ such that

$$\begin{aligned} &\max_{q, G_2} \quad \gamma \\ &\text{s.t.} \\ &\begin{bmatrix} \alpha Z_0 G_1 & 0 & G_1^\top Z_1^\top & \gamma Z_0 G_1 \\ 0 & qI & G_2^\top Z_1^\top & 0 \\ Z_1 G_1 & Z_1 G_2 & Z_0 G_1 & 0 \\ \gamma Z_0 G_1 & 0 & 0 & qI \end{bmatrix} \succeq 0, \\ &q > 0, \quad Z_0 G_2 = 0, \quad U G_2 - qK = 0, \end{aligned} \tag{17}$$

which will result in stability of the system (10) in exponential behaviour. □

Proof 3: See ref. [15, Section IV. B]. ■

The result of theorem 2 allows to maximize γ over the variables G_2 and q. The result of the theorem imply that any $\gamma \in [0, \gamma^\star[$ stabilizes the system, where $\gamma^\star$ is the solution for (17).

Now, we have all the components put together. A detailed algorithm for the entire process can be highlighted in algorithm 1.

Algorithm 1 Koopman-inspired Event-triggered State Feedback Control

Require: α, X_0, X_1, and Γ_0

1: Lift X_0, and X_1 via (7 b, and c)
2: Solve for Q_1 in the LMI given in (14)
3: Solve for the controller gain K in (15)
4: Maximize the threshold parameter γ to get $\gamma^\star$ in (17)
5: Choose any $\gamma \in [0, \gamma^\star[$, (typically the max. value gives wider inter-event time window)
6: **return** $\gamma^\star$, and K

IV. Illustrative Simulations and Results

A. Illustrative Example

We consider a case of nonlinear system with slow manifold inspired by an example from [34]:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} \rho x_1 \\ \kappa x_2 + (\rho^2 - \kappa) x_1^2 + u \end{bmatrix} \tag{18}$$

In this scenario, there exists a polynomial stable manifold defined as $x_2 = x_1^2$. Within the Koopman-inspired framework, if the correct lifting observable functions were chosen such that $\Xi(x) = \begin{bmatrix} x_1, & x_2, & x_1^2 \end{bmatrix}$, the nonlinear system in (18) can be expressed linearly as

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}_{k+1} = \begin{bmatrix} \rho & 0 & 0 \\ 0 & \kappa & (\rho^2 - \kappa) \\ 0 & 0 & \rho^2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}_k + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u_k \tag{19}$$

Considering the parameters, $\rho = 0.6$, and $\kappa = 1.2$ for the system, the eigenvalues of the system are $\lambda_1 = 0.6$, $\lambda_2 = 1.2$, and $\lambda_3 = 0.36$. Since $\lambda_2 > 1$, the system exhibits instability, and the goal is to stabilize it to the origin. We collected the data for $T = 45$ is enough for assumption 1 to hold – on a theoretical note, $T \geq m + p$ samples should be enough (i.e. in this example $T \geq 4$) to obey assumption 1. Therefore, $T = 4$ should work. The input signal is drawn from a normal distribution following $u \sim \mathcal{N}(0, 1)$.

Then, after deploying the steps in algorithm 1[2], we obtain $K = \begin{bmatrix} 0.0206 & -1.1109 & -0.1530 \end{bmatrix}$, which in turn gives $\gamma^\star = 0.7664$. We simulated the system for both ETC, and TTC and illustrated the behaviour in Fig. 2. All the results depicted in Fig. 2, are acquired after pulling the states back from the higher dimensional space, at this case from $\mathbb{R}^3$ to $\mathbb{R}^2$, by applying (8b) with $C = \left[\begin{smallmatrix} I_2 & 0 \\ 0 & 0 \end{smallmatrix}\right]$.

Fig. 2(a), illustrates the state evolution against time x_1 and x_2 under both ETC and TTC techniques. The trajectories for ETC demonstrate excellent tracking performance in comparison with the nominal TTC. This highlights the efficacy of the developed event-triggered approach in holding system stability while minimizing unnecessary updates.

Also, in Fig. 2(b), the graph shows that $\|e_k\|$ remains consistently below $\gamma\|x_k\|$, satisfying the triggering condition. As shown in Fig. 2(c), the substantial reduction in communication instances (40%) addresses potential concerns regarding communication overhead in practical implementations. However, that increased the control cost of the ETC as compared to the TTC. So we trade the communication resources efficiency with control cost. On another note, both Koopman based linearization ETC and TTC have control cost much lower than the nominal Taylor linearization technique, consistent with the results of Brunton et al. [34].

Additionally, results in Fig. 2(d) depicts the ratio of the Lyapunov function $V(x_{k+1})/V(x_k)$ compared to the stability decaying rate α. The ratio remains below α, verifying that the proposed event-triggered control law ensures exponential stability of the system. This gives another certificate of the theoretical guarantees provided by the Lyapunov-based stability analysis.

Finally, the results in Fig. 2(e) shows the trajectory of the system in the state space and the instances on the trajectory at which and event happens. Taking a closer look at the results, one can see that most of the events happen at the equilibrium position. That entails that the controller tries to keep the system around the origin, due to the instability nature of the system, the system deviates form the origin almost all the time. The Event-triggered based controller tries to compensate that behaviour with max inter-event timing according to what is depicted in Fig. 2(c).

[2]The used code can be found at: https://github.com/zmanaa/kinetc

V. Conclusion

To sum up, this study suggests an event-triggered control approach based on data-driven methods for discrete-time nonlinear systems. By lifting the nonlinear dynamics into a higher-dimensional linear representation inspired by the KO theory, the method makes it possible to create an event-triggered controller that is driven by data. Through the development of a closed-loop system and the implementation of a triggering strategy, the suggested method stabilizes the plant with less frequent control updates.

The event-triggered closed-loop system's exponential stability is guaranteed by the stability analysis, which is based on the Lyapunov criterion. Numerical simulations and theoretical analysis are used to show how effective the suggested strategy is. This work creates opportunities for real-world application in networked control systems and advances event-triggered control techniques for nonlinear systems.

Appendix

Lyapunov Stability Analysis

Here we present the exponential stability of the system given by

$$z_{k+1} = Z_1 L z_k + Z_1 N e_k \tag{20}$$

in a Lyapunov sense with the candidate $V(k) = z_k^\top S z_k$. First, we compute $V(z_{k+1})$

$$\begin{aligned} V(z_{k+1}) &= (Z_1 L z_k + Z_1 N e_k)^\top S (Z_1 L z_k + Z_1 N e_k) \\ &= z_k^\top L^\top Z_1^\top S Z_1 L z_k + z_k^\top L^\top Z_1^\top S Z_1 N e_k + \dots \\ &\dots e_k^\top N^\top Z_1^\top S Z_1 L z_k + e_k^\top N^\top Z_1^\top S Z_1 N e_k \end{aligned} \tag{21}$$

Then, we reach to Lyapunov exponential stability condition with convergence rate α by defining $V(z_{k+1}) \leq \alpha V(z_k)$. We evalute the identity based on the candidate Lyapunov function and we get

$$\begin{aligned} & z_k^\top L^\top Z_1^\top S Z_1 L z_k + z_k^\top L^\top Z_1^\top S Z_1 N e_k + \dots \\ & \dots e_k^\top N^\top Z_1^\top S Z_1 L z_k + e_k^\top N^\top Z_1^\top S Z_1 N e_k \leq \alpha z_k^\top S z_k \end{aligned} \tag{22}$$

Afterwards, by defining $v = \left[\begin{smallmatrix} z_k \\ e_k \end{smallmatrix}\right]$, Eqn. (22) can be written in the form of $v^\top \Psi v \leq 0$, where

$$\Psi = \begin{bmatrix} L^\top Z_1^\top S Z_1 L - \alpha S & L^\top Z_1^\top S Z_1 N \\ N^\top Z_1^\top S Z_1 L & N^\top Z_1^\top S Z_1 N \end{bmatrix} \tag{23}$$

Therefore, the Lyapunov stability condition for the system can be written as

$$\begin{bmatrix} z_k \\ e_k \end{bmatrix}^\top \begin{bmatrix} L^\top Z_1^\top S Z_1 L - \alpha S & L^\top Z_1^\top S Z_1 N \\ N^\top Z_1^\top S Z_1 L & N^\top Z_1^\top S Z_1 N \end{bmatrix} \begin{bmatrix} z_k \\ e_k \end{bmatrix} \leq 0 \tag{24}$$

If condition 24 is satisfied, it then guarantees exponential stability of the system with convergence rate α.

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